computational game theory

com810s

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extensive form games

intro

- dynamic description of game
- sequential actions by players
- representation as a game tree (graph theory)
  - root node
  - leaves represent outcomes
  - intermediate nodes are decision nodes (choice) for players

- P: set of players
- A: set of all actions
- H: set of decision nodes
- Z: set of terminal nodes (leaves)

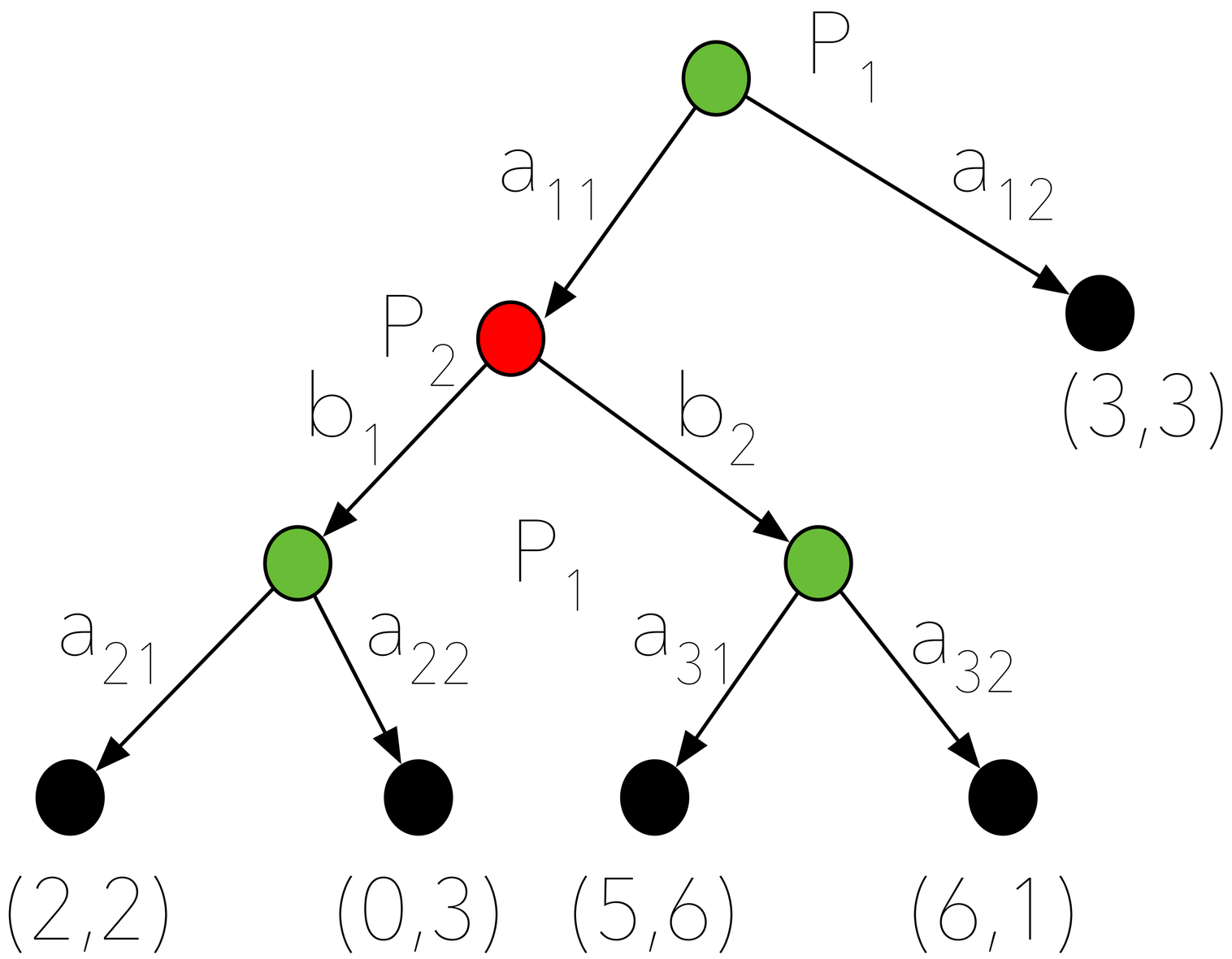
$$\mathscr{G} = \langle \mathsf{P}, \mathsf{A}, \mathsf{H}, \mathsf{Z}, \chi, \rho, \sigma, \mathrm{u} \rangle$$

$$\chi : \mathsf{H} \to 2^{\mathsf{A}}$$

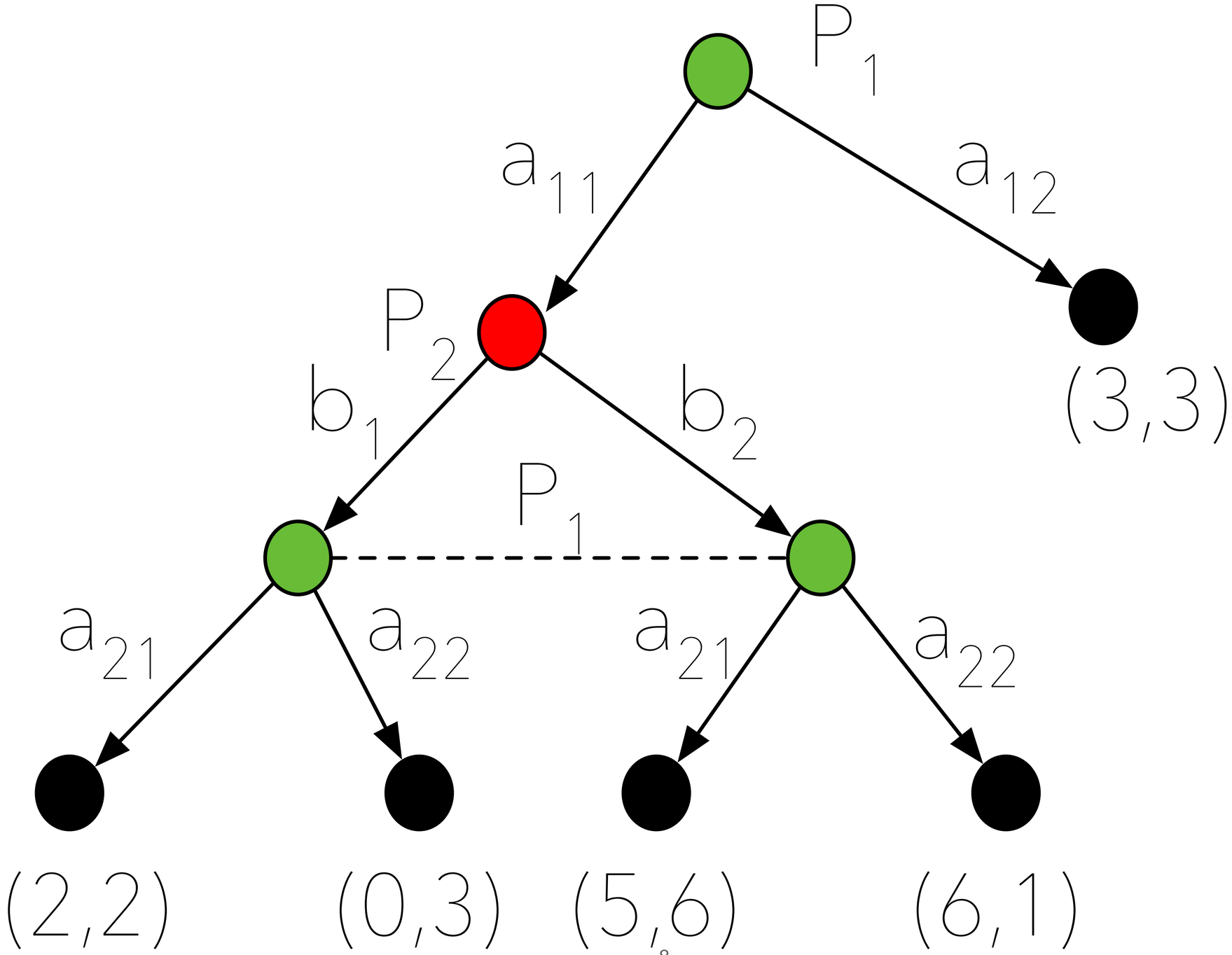
$$\rho : \mathsf{H} \to \mathsf{P}$$

$$\sigma : \mathsf{H} \times \mathsf{A} \to \mathsf{H} \cup \mathsf{Z}$$

$$\mathrm{u} : \mathsf{z} \to \mathbb{R}$$


P1
a11
a12
P2
b1
b2
(3,3)
P1
a21
a22
a31
a32
(2,2)
(0,3)
(5,6)
(6,1)

- strategy set
  - pure strategies: actions from the information set
    - cartesian product of decision nodes (X)
- information set
  - subset of decision nodes from same player with similar actions
  - perfect or imperfect information (define the information set)
  - can use information set to mimic simultaneous moves in a sequential game
- strategy vs action
  - Strategy: function returning the list of branches to be considered in the tree


P1
a11
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b1
b2
P1
a21
a22
a21
a22
(2,2)
(0,3)
(5,6)
(6,1)

- strategy set
  - previous game
    - case 1: perfect information
    - case 2: imperfect information

- strategy set
  - perfect information
    - P1 = $\{(a_{11}, a_{21}, a_{31}), (a_{12}, a_{21}, a_{31}), (a_{11}, a_{22}, a_{31}), (a_{12}, a_{22}, a_{31}), (a_{11}, a_{21}, a_{32}), (a_{12}, a_{21}, a_{32}), (a_{11}, a_{22}, a_{32}), (a_{12}, a_{22}, a_{32})\}$
    - P2 = $\{b_1, b_2\}$
  - imperfect information
    - P1 = $\{(a_{11}, a_{21}), (a_{12}, a_{21}), (a_{11}, a_{22}), (a_{12}, a_{22})\}$
    - P2 = $\{b_1, b_2\}$
  - pure strategies in imperfect information games select actions from the information sets

- represent the game in induced normal form

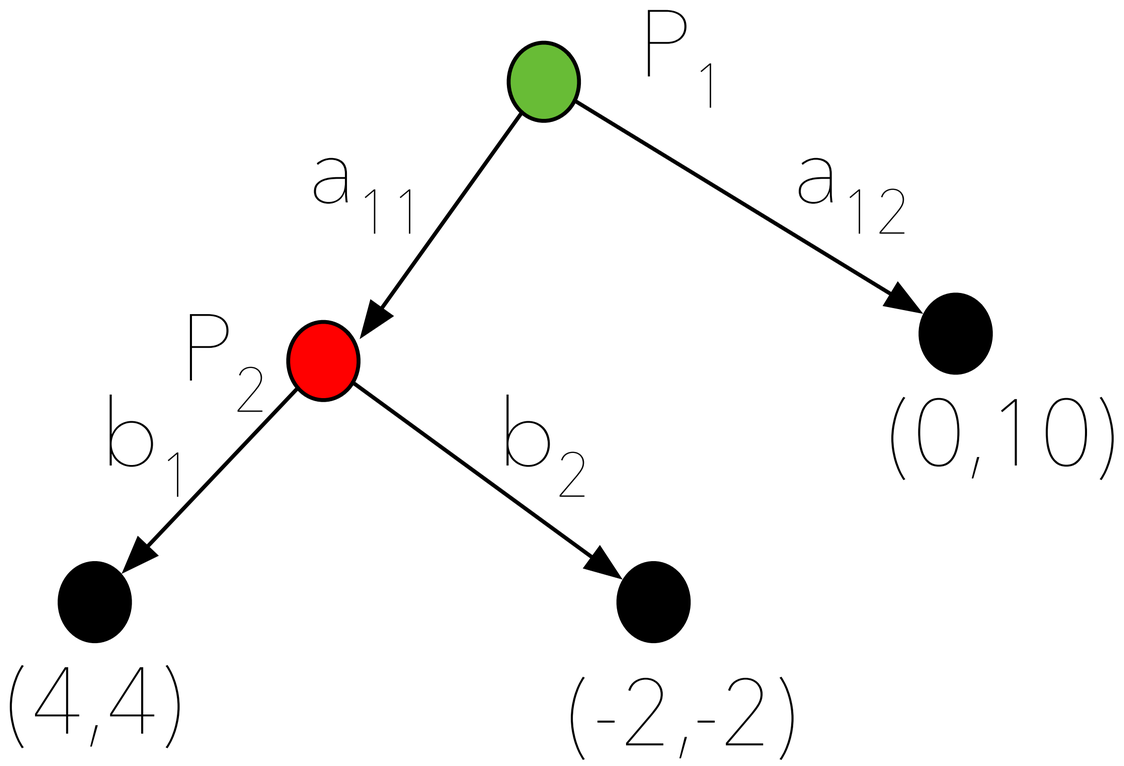
solution concepts

- rules
  - every node in the game tree is a successor to the initial node
  - every node has exactly one immediate predecessor
  - branches from the same node have different actions
  - information set is related to one player and same actions

subgame perfect equilibrium

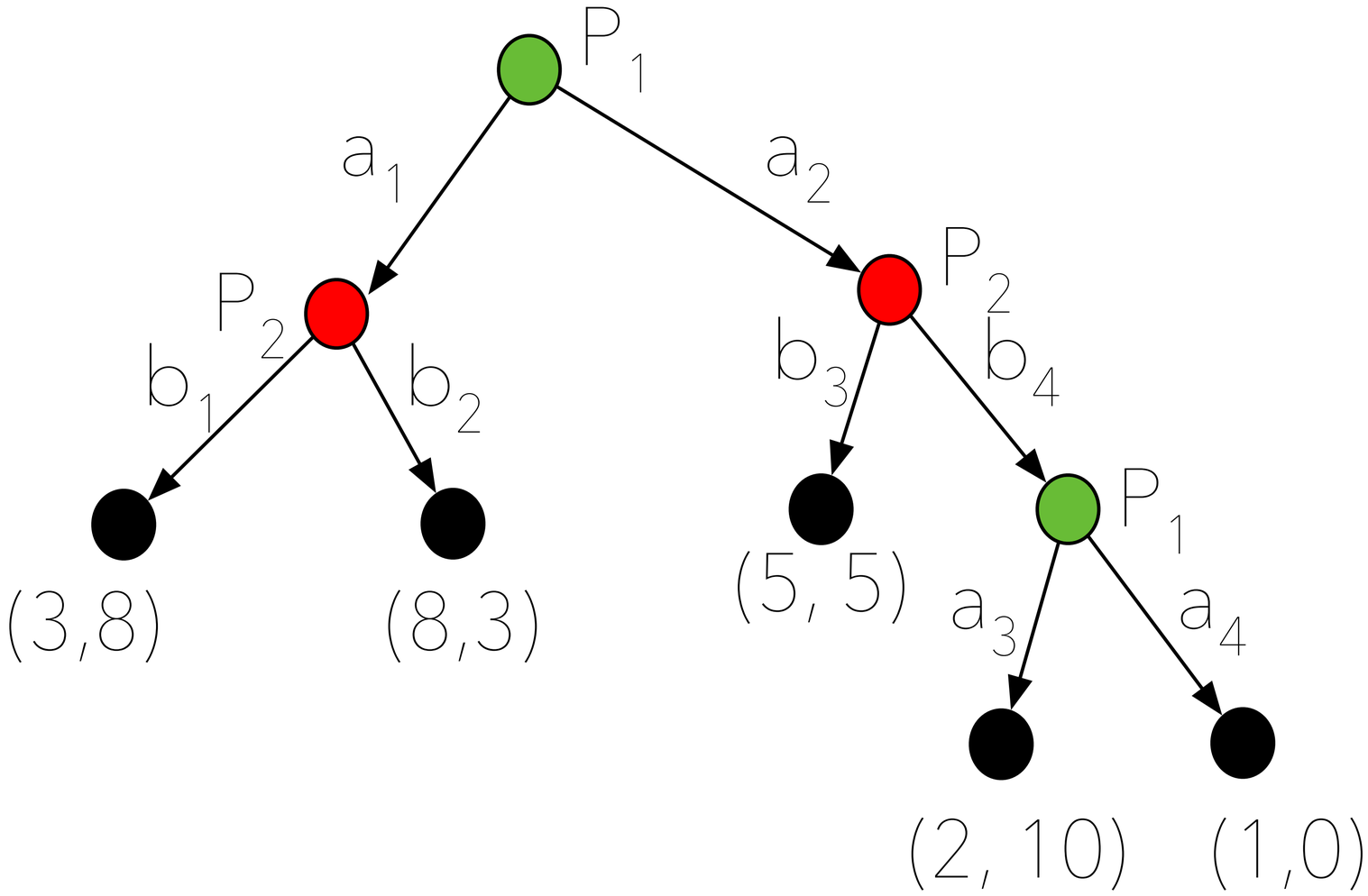
- challenge with NE

- consider the game on the right-hand side
- find the NE
- are all the NE rational?



- the game has two PSNE
  - $\{\{a_{11}, b_1\}, \{a_{12}, b_2\}\}$
  - however, $\{a_{12}, b_2\}$ is not **sequentially rational** since P_2 cannot rationally choose b_2 once P_1 chooses a_{11}
  - b_2 represents a threats for P_2

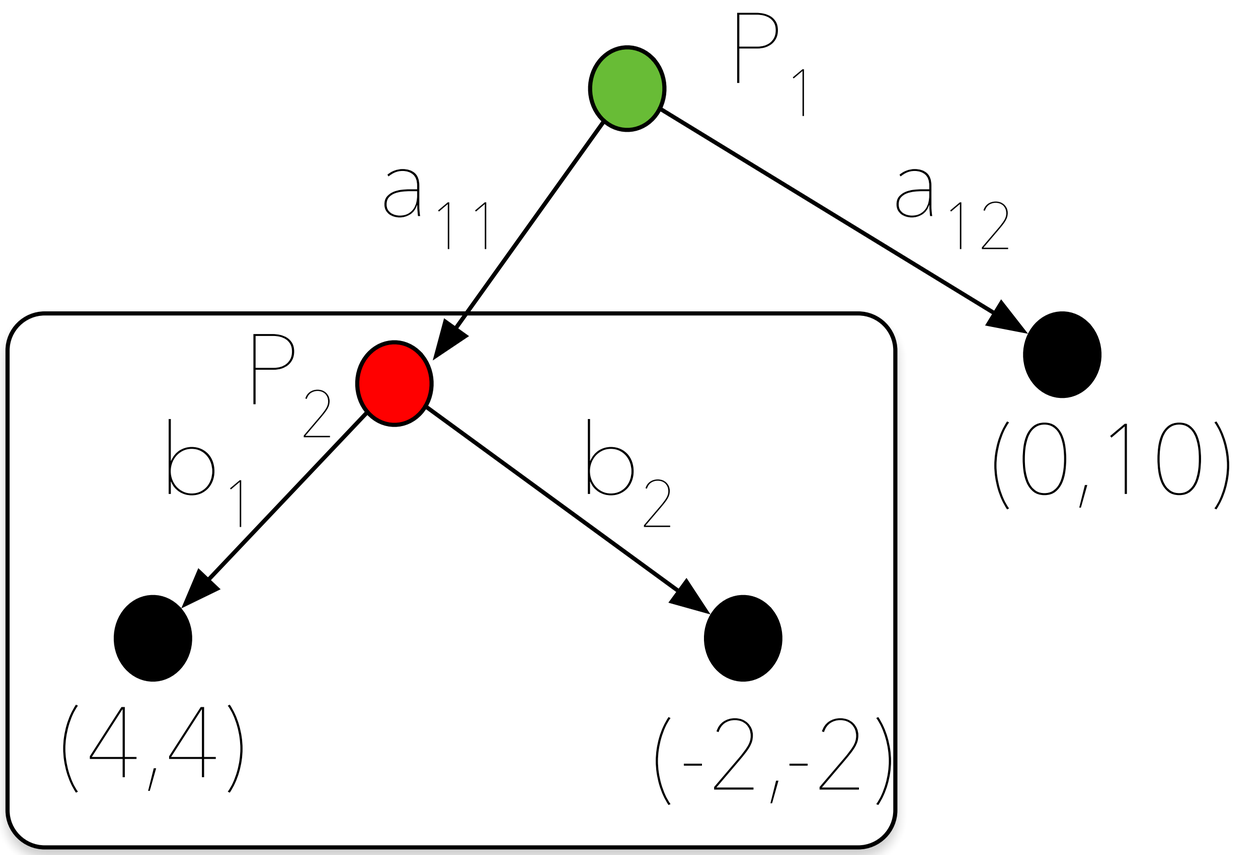
- what about this game?



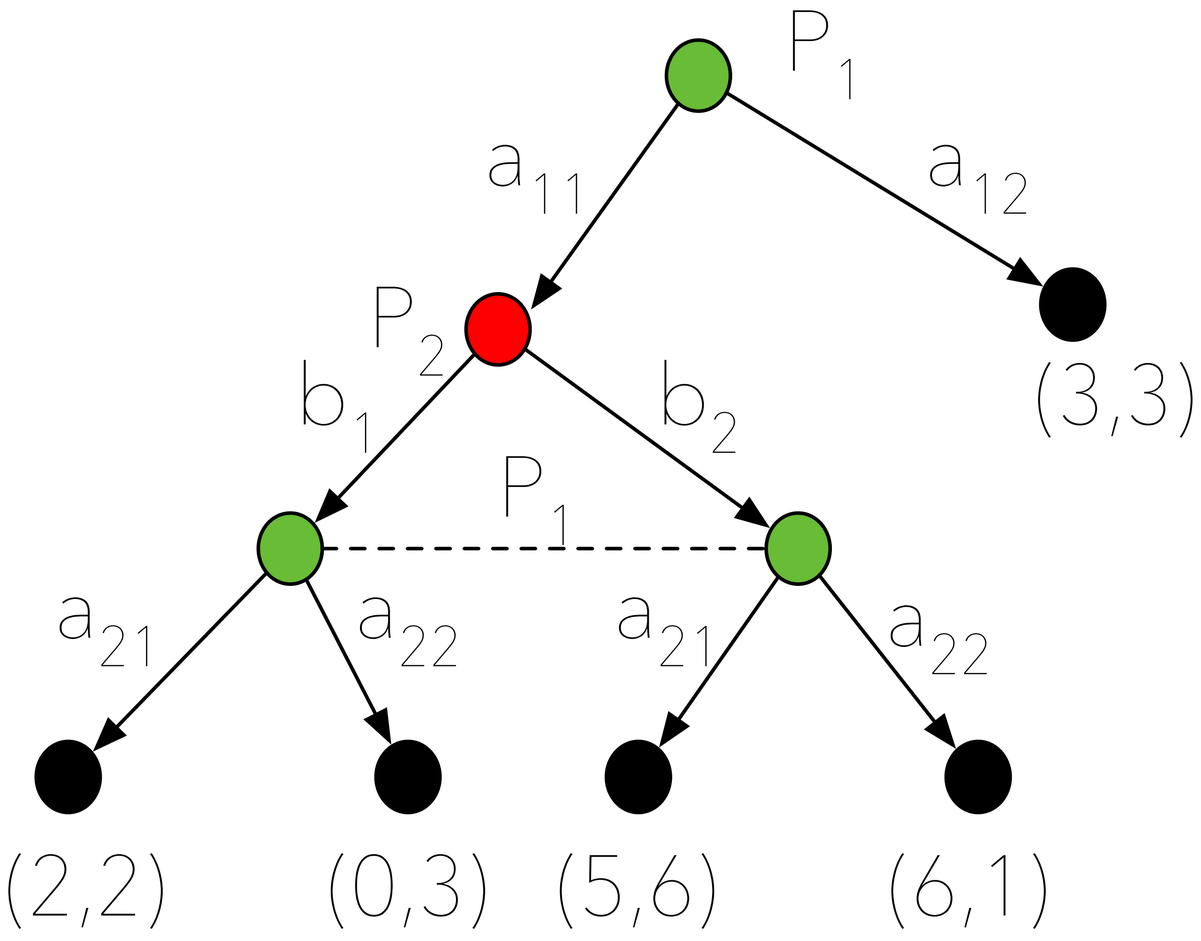
- the game has three PSNE
  - $\{\{a_1a_3, b_1b_4\}, \{a_1a_4, b_1b_4\}, \{a_2a_4, b_1b_3\}\}$
  - however, $\{a_2a_4, b_1b_3\}$ is not **sequentially rational**; a_4 represents a threat to P_2

subgame pefect equilibrium

- subgame
  - tree structure defined by a node and its subtree
  - a subgame should not split/break an information set

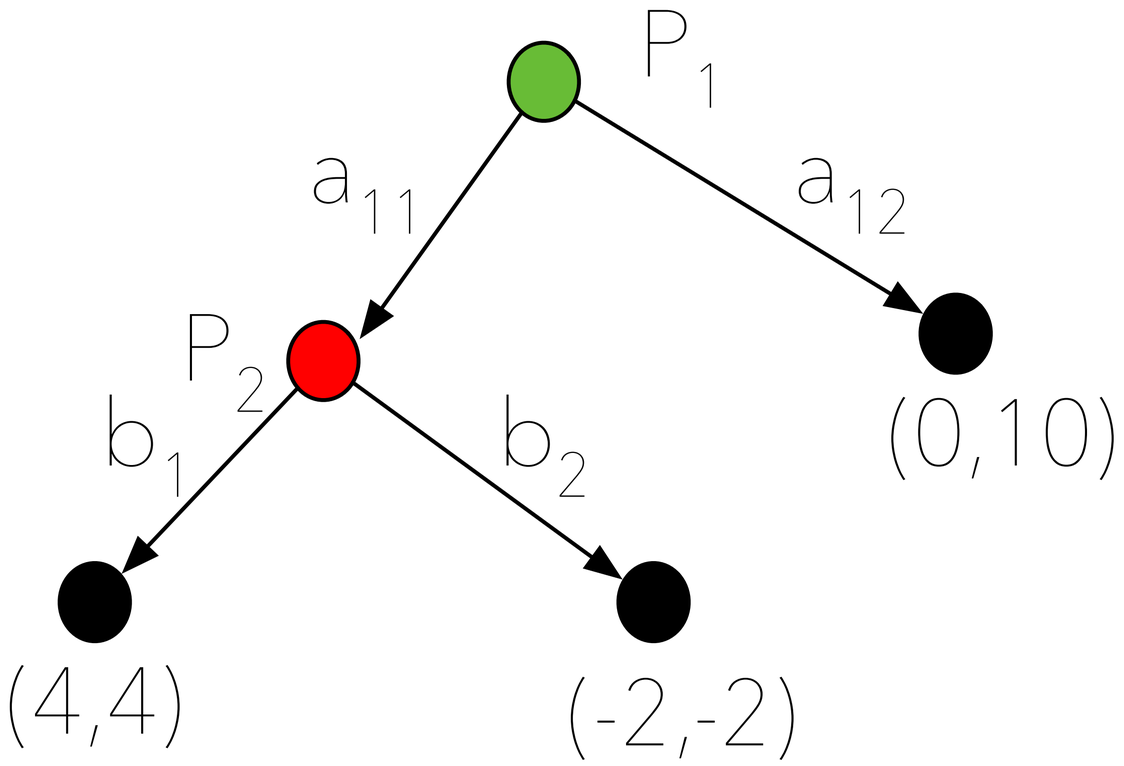

P1
a11
a12
P2
b1
b2
(0,10)
(4,4)
(-2,-2)

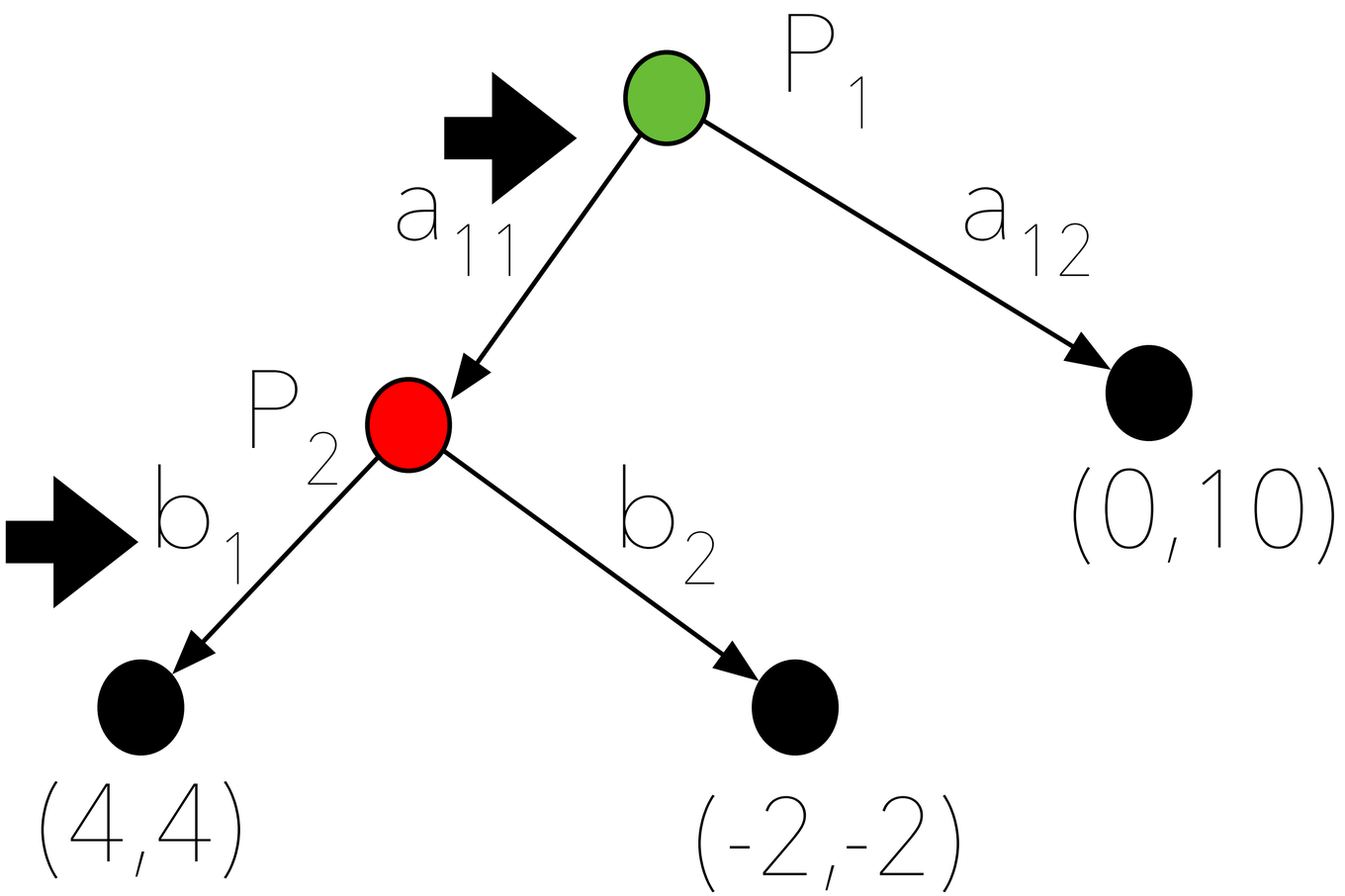
- list the subgames in right-hand side games?



- strategy profile is a subgame perfect equilibrium, if it specifies a NE in each subgame
- method: backward induction
  - from the leaves, traverse the tree selecting the branch that yields the highest payoff

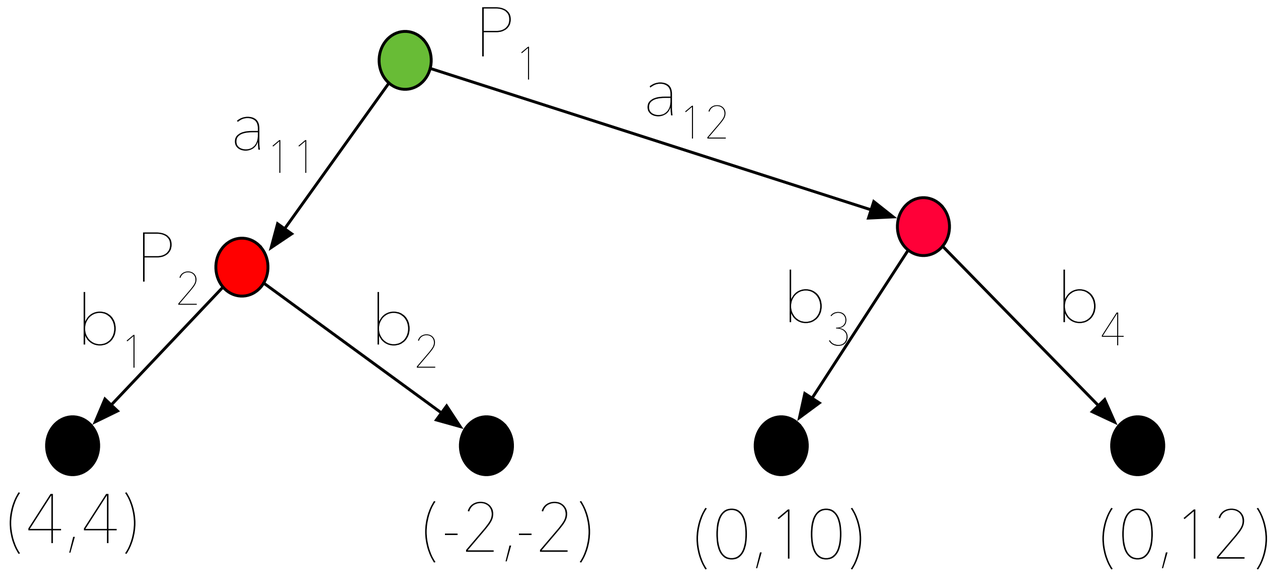
- backward induction illustration

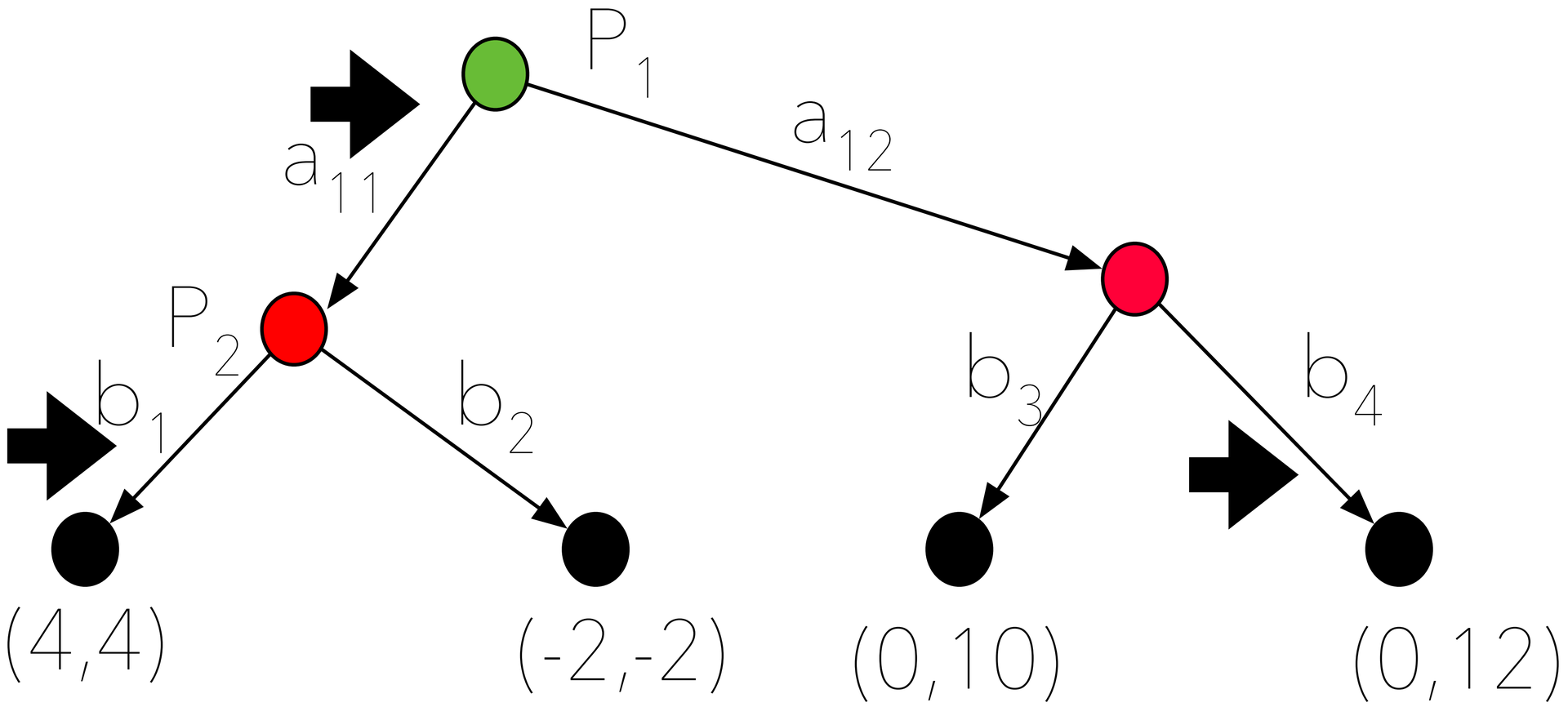



P1
a11
a12
P2
b1
b2
(0,10)
(4,4)
(-2,-2)

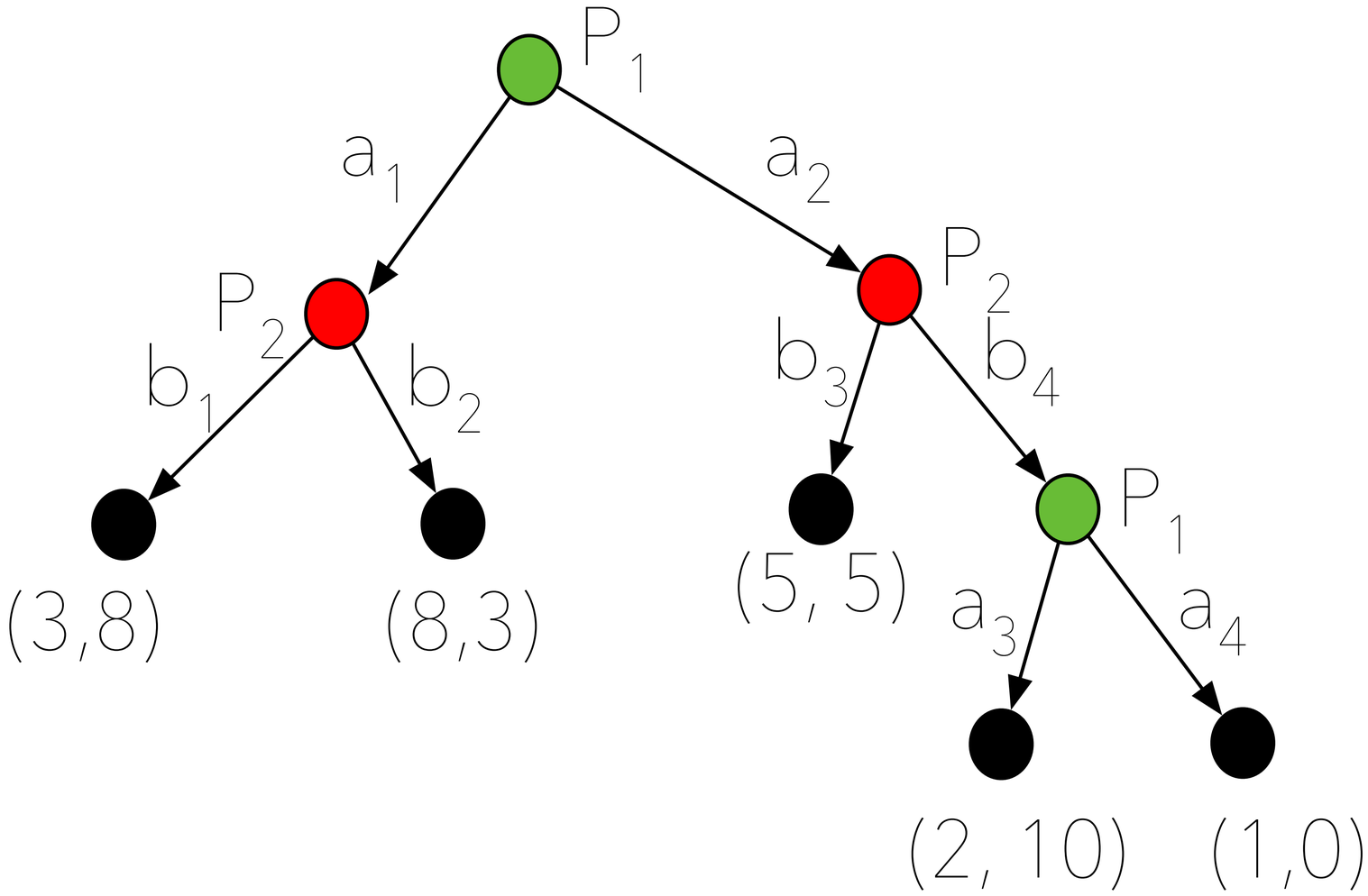
- every SPE is an NE; however every NE is not an SPE

- what about this game?




P1
a11
a12
P2
b1
b2
b3
b4
(4,4)
(-2,-2)
(0,10)
(0,12)

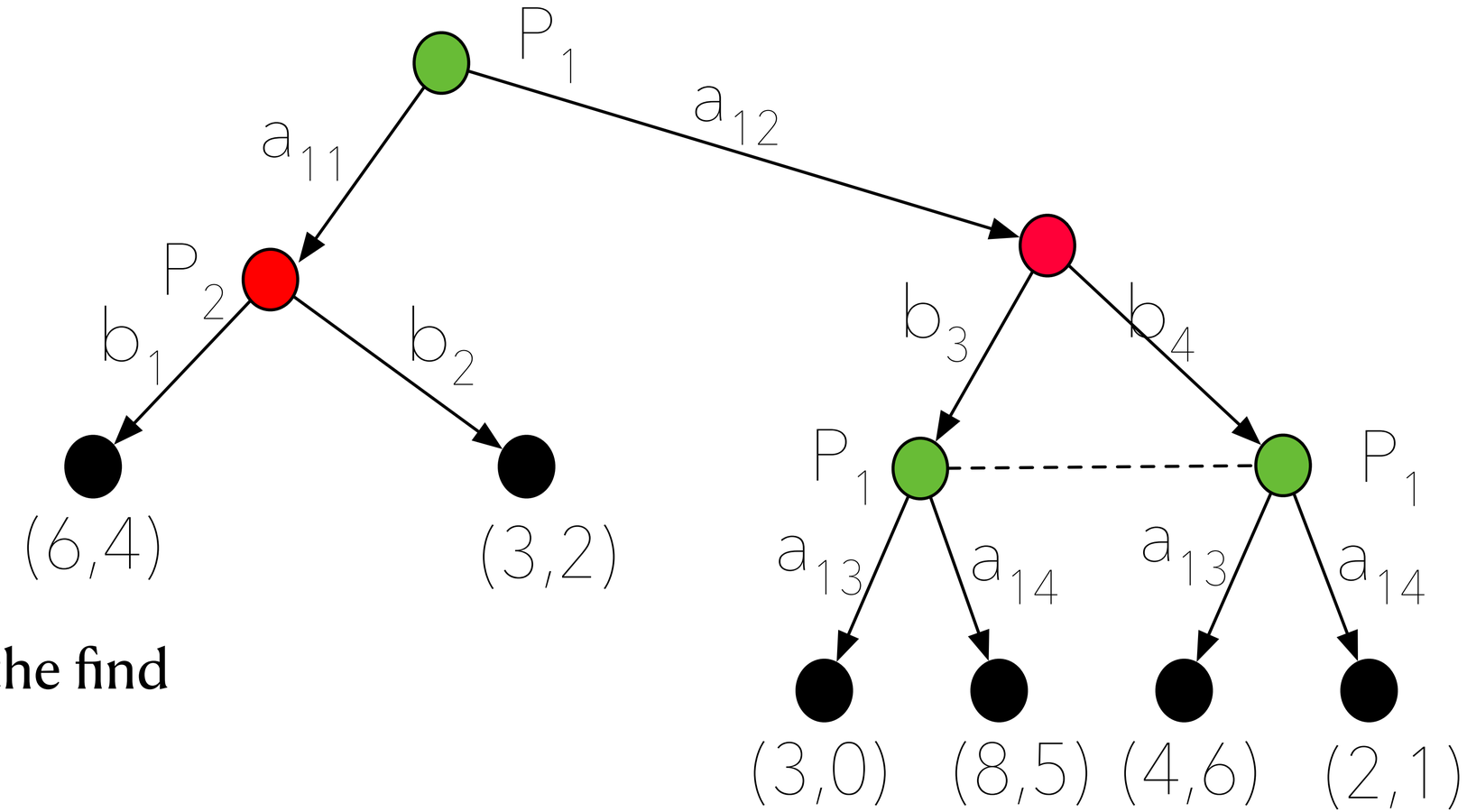
- and this?



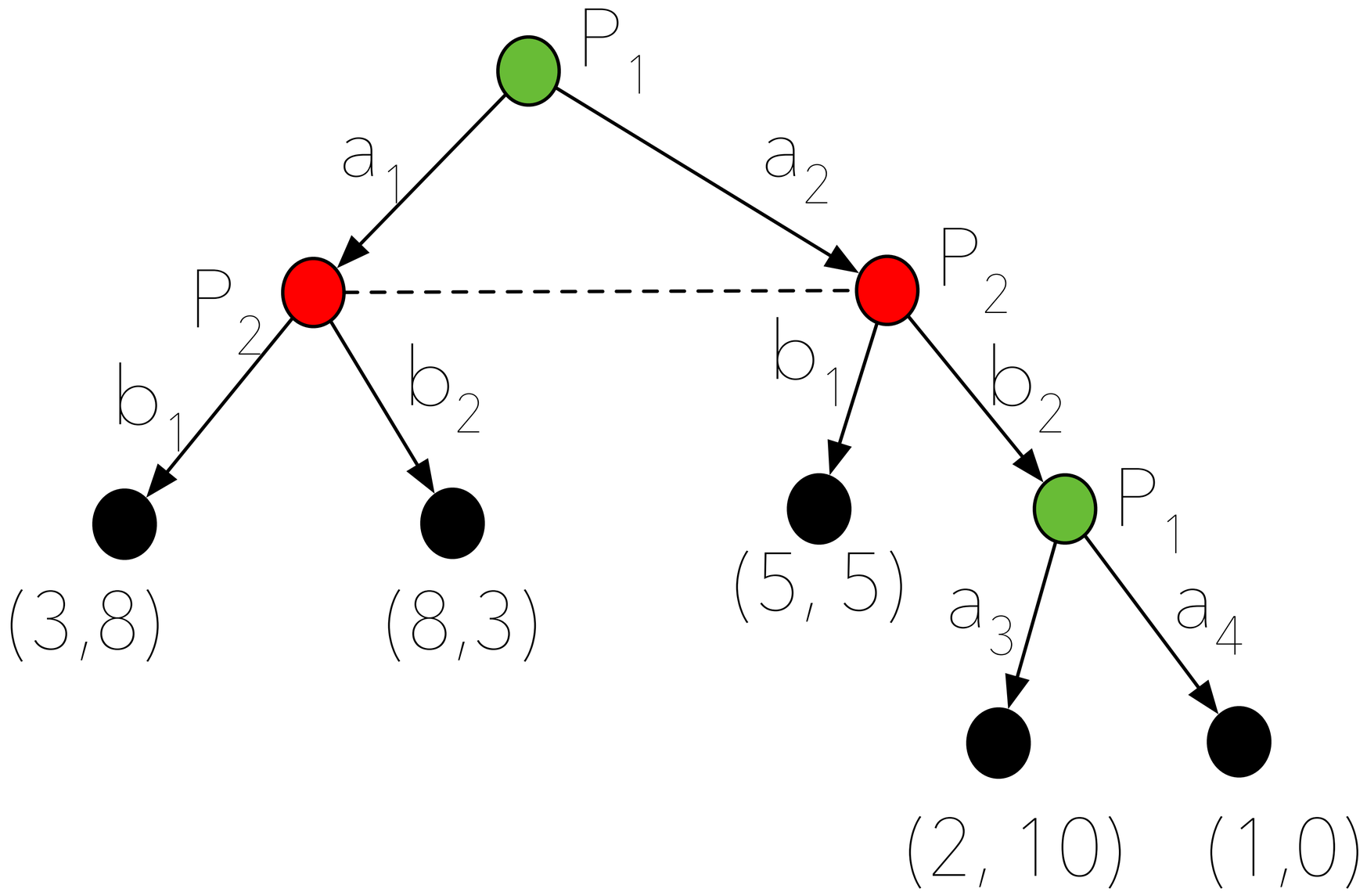
- backward induction — summary
  - if node is terminal, return payoff
  - else, recursively explore the branch to find best action

imperfect information games

- find the subgame perfect equilibrium for this game
  - hint
    - list the subgames and the find the NE for each



- SPNE
  - $\{\{a_{11}a_{13}, b_1b_4\}, \{a_{12}a_{14}, b_1b_3\}\}$


P1
a1
a2
P2
P2
b1
b2
b1
b2
P1
(3,8)
(8,3)
(5, 5)
a3
a4
(2, 10)
(1,0)

- information sets represent partitions over H

$$\mathscr{G} = \langle \mathsf{P}, \mathsf{A}, \mathsf{H}, \mathsf{Z}, \chi, \rho, \sigma, \mathrm{u}, \mathtt{I} \rangle$$

$$\mathtt{I}_{\iota} = \{\mathrm{h} \in \mathsf{H} \cdot \rho(\mathrm{h}) = \mathrm{P}_{\iota}\};$$

$$(\mathrm{h} \in \mathtt{I}_{\iota,J} \wedge \mathrm{h}' \in \mathtt{I}_{\iota,J}) \implies \chi(\mathrm{h}) = \chi(\mathrm{h}') \wedge \rho(\mathrm{h}) = \rho(\mathrm{h}')$$

- mixed vs behavioural
  - mixed strategy: probability distribution over pure strategies
    - a_1a_3: 0.6 and a_2a_4: 0.4
    - randomisation over all pure strategies
  - behavioural strategy: different probability distribution for each information set
    - a_1: 0.5; a_3: 0.3
    - randomisation at every information set
    - probability distribution over the “out-egdes” of an information set

- game with perfect recall
  - each player remembers the action it took and the information it had
- Kuhn's theorem
  - in perfect recall games, behavioural strategy is "output equivalent" to mixed strategy

- belief system
  - given an information set I owned by player P_i, choose a probability distribution over the vertices in I

$$\sum_{v \in I} \mu_\ell(v) = 1$$

- assessment (β, μ)

v_0, v with path $(\mathrm{v}_0, \mathrm{v}_1, \ldots, \mathrm{v}_{k-1}, \mathrm{v})$ $\beta(\mathrm{v}_0, \mathrm{v}) = \beta(\mathrm{v}_0, \mathrm{v}_1)\beta(\mathrm{v}_1, \mathrm{v}_2)\ldots\beta(\mathrm{v}_{k-1}, \mathrm{v})$

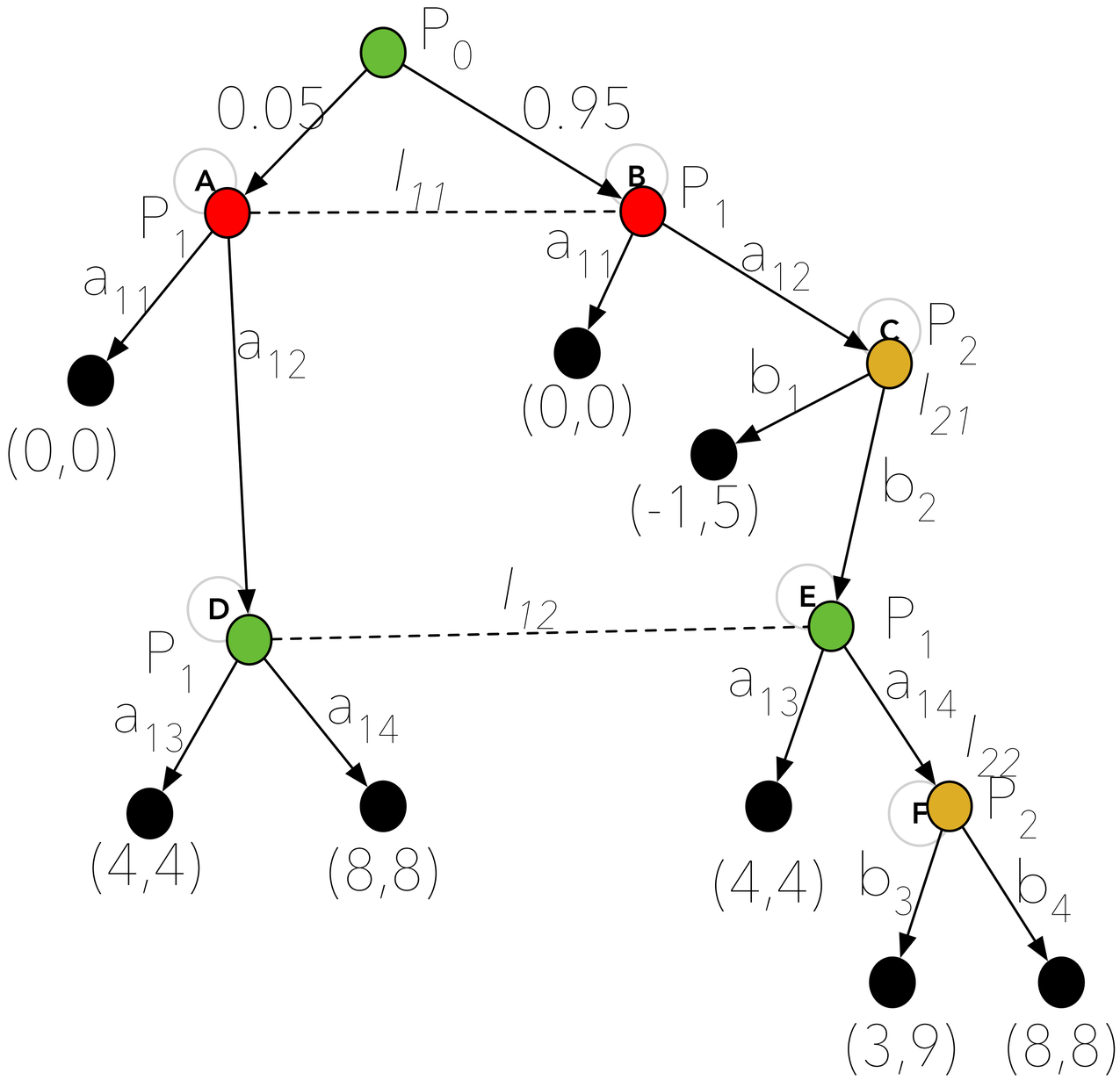
- if a vertex is owned by a chance node, use the given probabilities
- the probability of an information set is the sum of the probabilities of its vertices

- a belief system is weakly consistent with a behavioural strategy profile if

$$\beta(I) > 0 \wedge \mathrm{v} \in I \implies \mu(\mathrm{v}) = \frac{\beta(\mathrm{v})}{\beta(I)}$$

- given an extensive game with perfect recall
  - an assessment represents a weak sequential equilibrium
    - the belief system is weakly consistent with the strategy profile
    - the expected payoff on any information set is the maximum expected utility

- example
  - consider the game below
  - find a sequential equilibrium


P0
0.05
0.95
A
B
P1
I11
P1
a11
a12
a11
a12
C
P2
(0,0)
(0,0)
b1
I21
(-1,5)
b2
D
I12
E
P1
P1
a13
a14
a13
a14
I22
F
P2
(4,4)
(8,8)
(4,4)
b3
b4
(3,9)
(8,8)

- solution (1)
  - find a consistent assessment: **behavioural strategy** and **consistent belief**
  - note
    - P1 has two information sets (I_{11} and I_{12})
    - P2 has two information sets (I_{21} and I_{22})
  - let
    - p_1: probability for a_{12}
    - p_2: probability for b_2
    - p_3: probability for a_{14}
    - p_4: probability for b_4

- solution (2)
  - can determine p_4 using backward induction
  - on information set I_{12}, find consistent belief for reaching node E (node D's belief can be inferred)
  - equate expected utility for P1 for actions a_{13} and a_{14} and analyse options for p_3
  - solve expected utility equation for P_2 at I_{21}
  - Evaluate the consistency of behavioural probabilities and belief
  - in case of consistent combinations, find the rest of the equilibrium

summary

- game tree to represent sequential behaviour
- complete information
  - backward induction: subgame perfect equilibrium
- incomplete information
  - simultaneous moves: strategic form (NE)
  - behavioural strategy and sequential equilibrium

zero-sum games

- sum of payoffs in each case is **zero**

$$\mathsf{M} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}; \quad \mathsf{N} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

		P_2	
		b_1	b_2
P_1	a_1	1	-1
	a_2	-1	1

- security levels
  - u1: player 1 and u2 for player 2

$$\bar{u}_1 = \max_{a \in A_1} \min_{a' \in A_2} u(a, a')$$

$$\bar{u}_2 = \min_{a' \in A_2} \max_{a \in A_1} u(a, a')$$

- security strategy
  - action or strategy consistent with security level

- define security levels and strategies for this game

		P_2	
		b_1	b_2
P_1	a_1	3	0
	a_2	1	2

- is there a PSNE in this game?
- what about MSNE?

		P_2	
		b_1	b_2
P_1	a_1	1	-1
	a_2	-1	1

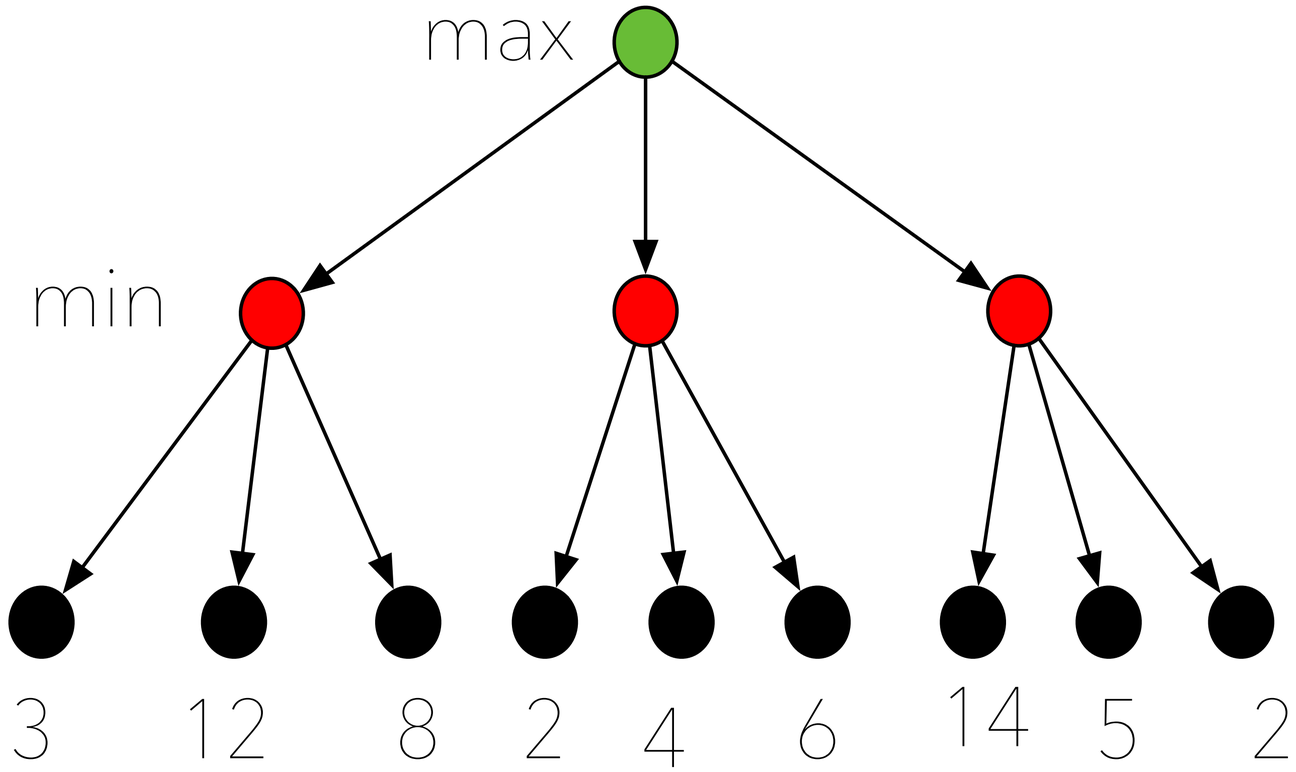
- is there a PSNE in this game?
- what about MSNE?

		P_2	
		b_1	b_2
P_1	a_1	3	1
	a_2	5	4

- note
  - saddle point: when security levels coincide
  - yields a PSNE

- minimax tree search
  - explore game tree with min and max players and find solution

- consider the game on the rhs
  - find the solution using the minimax algorithm



- rules
  - terminal node
    - return utility
  - max node
    - evaluate all nodes and return max
  - min node
    - evaluate all nodes and return min

- see alpha-beta algorithm for efficient tree traversal